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Predictive Modelling of Top Indian Banks' Stock Prices using NLP, Deep Learning and FINBERT

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Abstract- Stock price prediction remains a complex challenge due to the dynamic and nonlinear nature of financial markets. Traditional statistical methods often struggle to capture temporal dependencies and external influences such as investor sentiment and financial news. This study proposes a sentiment-aware stock prediction framework for forecasting stock prices of top Indian banking sector companies using Natural Language Processing (NLP), FinBERT sentiment analysis, and deep learning models. Historical stock market indicators such as open, high, low, close prices, and trading volume are combined with sentiment scores extracted from financial news data. Sequential neural networks, including Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) models, are employed to capture temporal dependencies and complex market patterns. Model performance is evaluated using metrics such as MAE, RMSE, and R². The proposed system integrates sentiment-driven insights with historical price analysis to improve prediction reliability and support intelligent financial decision-making.

1. INTRODUCTION

Stock markets play an important role in the economy of any country. They help companies raise money by offering stocks on a stock exchange. They also help people earn money by earning on the appreciation of their investments. Stock prices also help to determine the economic strength and stability of any country. They also help investors get information about their investments in relation to the general economy. Since stock prices are affected by a large number of different variables, it can be very difficult to predict how they will behave in the future. These variables include: how companies are performing, how the government is performing, how the world is performing in general, how investors feel about companies in general, and any information that is released to the general public about companies. Due to the large amount of unknowns about how all these different variables will affect stock prices in the future, it can be very difficult to correctly predict stock prices with the help of financial analytics.

Stock Price Influences

- Legal Factors (L):** No legal factors
- Political Factors (P):** No political factors
- Environmental Factors (E):** No environmental factors
- Technological Factors (T):** No technological factors
- Economic Factors (E):** Economic factors
- Social Media (S):** Social media sentiment

Fig. 1.1 Various factors influencing stock price movements

Investors have traditionally used both technical and fundamental analysis techniques to predict the prices of stocks. Technical analysis is based on historically collected market data such as prices and the amount of trading, while fundamental analysis is based on various economic indicators, publicly available financial information from individual companies, and economic trends in the industry as a whole. Despite their widespread use by investors, traditional financial analysis techniques fail to account for complex and nonlinear relationships that exist within financial market time series data. This failure of traditional analysis techniques has prompted researchers to increasingly investigate the use of deep learning and machine learning for purposes of stock market price prediction. Due to the rapid innovation in the field of artificial intelligence, eg. deep learning techniques now exist, which can model very complex patterns learned from very large datasets [2]. For example, Long Short Term Memory (LSTM) and Gated Recurrent Unit (GRU) models are both types of deep learning models that have been shown to effectively model sequential data that are related to a time series (timestamped), such as stock prices and to find temporal dependencies in the movement of stock prices [1][24]. In addition to using historical market price data, research has shown that the sentiment of investors that is derived from financial news articles and social media makes a significant contribution to trend changes in the markets [6]. Recently developed Natural Language Processing models, such as FinBERT, have been created to assist users in analysing the financial text and extracting sentiment from that text [10]. By integrating sentiment analysis, which can be performed using FinBERT, with historical stock data, and with deep learning methods for the prediction of stock prices, it will be possible to create cleaner, more reliable stock price prediction systems.

AI-Based Stock Prediction Workflow

- Data Collection:** Gather open, close, high, and low stock prices
- Feature Engineering:** Collect relevant financial news articles
- Model Training:** Combine historical stock data and sentiment features
- Prediction:** Generate stock price predictions using the trained model

Fig. 1.2. Conceptual framework for sentiment-aware stock price prediction

The goal of this project is to construct a predictive modelling system by incorporating FinBERT sentiment analysis with various deep learning architectures, like an LSTM or GRU, and to predict stock prices of large companies in India's banking sector. The system will

combine quantitative metrics from corporate finance and qualitative metrics from the sentiment analysis, so that predictions made using the model will have better accuracy and provide valuable information to both investors and financial professionals.

II. LITERATURE SURVEY

The stock market prediction is an area that has been heavily researched due to its ability to help investors make informed financial decisions. Traditional forecasting methods such as regression models, Autoregressive Integrated Moving Average (ARIMA), and Generalised Autoregressive Conditional Heteroscedasticity (GARCH) have been used to analyse financial time-series data; however, these methods are often unable to model the non-linear and complex behaviours found in stock market data, thus prompting researchers to look at machine learning and deep learning methods [5][18]. Recent research has demonstrated that deep learning models such as Recurrent Neural Networks (RNN), Long Short Term Memory (LSTM) and Gated Recurrent Units (GRU) perform strongly in modelling sequential financial data [3][13][24]. An LSTM-GRU hybrid model is presented in this study to account for both the long-term dependencies and the short-term fluctuations in the movement of stock prices, leading to an improvement in the accuracy of predictions compared to standalone models. Many researchers have worked on the impact that sentiment analysis has on the prediction of the stock market. There is a lot of information available on the sentiments and attitudes of the investors towards the stock markets through social media sources and news agencies. By

analysing these sources of information, predictive modelling can use both subjective information on factors that would affect the stock prices and objective information in the form of stock price data to predict the stock prices. For example, in one such research paper, a model named Transductive Long Short-Term Memory (TLSTM) was used to predict the stock prices. The model used the sentiment information from the social media data and the stock price data to predict the stock price. The results showed that the integration of the sentiment information from the social media data with the financial data improved the accuracy of the predictions. Moreover, a study employed a model called FinBERT, which is a Natural Language Processing model for the financial industry, that helped derive the sentiment of the data from the social media platform, which was further employed to make a prediction of the stock prices by combining the data with the historical data using a Hybrid LSTM-GRU network. The study revealed that the model based on the sentiment data was more efficient compared to the traditional models based on numerical data. Research on bank stock has demonstrated that combining news sentiment analysis with deep learning models (LSTM and GRU) provides valuable insight into how a stock price will move [6][19]. For example, FinBERT was used to create a sentiment score from financial news articles, then combined with historical stock price data to create prediction algorithms from this data. The present study builds upon these works by developing a sentiment-aware predictive framework to estimate future stock prices of the largest banking companies in India.

III. COMPARISON

Ref	Authors	Technique Used	Dataset	Evaluation Metrics	Key Contribution
1	Farhadi et al.	Hybrid LSTM-GRU	Stock Market Data	MSE, RMSE	Combined LSTM and GRU to capture long and short dependencies, improving prediction accuracy
2	Peivandizadeh et al.	TLSTM + Sentiment Analysis	Stock + Social Media	Accuracy, RMSE	Integrated sentiment analysis with stock data using transductive learning
3	Girsang et al.	FinBERT + LSTM-GRU	Cryptocurrency Data	MSE, RMSE, MAE	Used FinBERT to extract sentiment from tweets for price prediction
4	Jeet et al.	LSTM + GRU + FinBERT	Banking Stocks	MAE, RMSE	Combined news sentiment with stock data for forecasting banking stocks
5	Behura et al.	Multi-layer LSTM	Financial Data	RMSE	Demonstrated effectiveness of LSTM for sequential financial prediction
6	Koo et al.	LSTM + Feature Clustering	Bitcoin Dataset	RMSE	Improved volatility prediction using feature clustering
7	Chen et al.	Feature Selection + LSTM	Financial Indicators	Accuracy	Introduced feature engineering for improving prediction reliability
8	Wu et al.	CNN + LSTM	Financial Forums	Accuracy	Combined investor sentiment from forums with stock prediction models
9	Swathi et al.	LSTM + Twitter Sentiment	Social Media Data	Accuracy	Used Twitter sentiment to improve stock price forecasting
10	Basant et al.	Sentiment + Deep Learning	Social Media Data	Accuracy	Applied sentiment classification with deep learning models
11	Harguem et al.	SVM Sentiment Model	NASDAQ Dataset	Accuracy	Investigated the correlation between Twitter sentiment and stock price
12	Gupta et al.	Machine Learning + Sentiment	Financial News	RMSE	Combined news sentiment with machine learning models
13	Zhang et al.	Wavelet + SVM	Stock Index	Accuracy	Reduced noise in financial datasets before prediction
14	Mahmoodi et al.	PSO + SVM	Financial Market Data	Accuracy	Optimised classification for stock price prediction
15	Kurani et al.	ANN	Financial Dataset	Accuracy	Demonstrated ANN robustness in stock prediction
16	Tan et al.	Neural Network Model	Stock Market Data	Accuracy	Developed neural networks for short-term prediction

17	Nauck et al.	Genetic Algorithm + ANN	Stock Market	Accuracy	Built decision support systems for stock trading
18	Yim et al.	ARMA / GARCH	Financial Series	RMSE	Compared econometric models for market volatility
19	Rodriguez et al.	ANN + Technical Indicators	Stock Data	Accuracy	Improved prediction using technical indicators
20	Sivadasan et al.	RNN / LSTM / GRU	Stock Time Series	RMSE	Compared deep learning architectures for forecasting

IV. PROPOSED METHODOLOGY

The goal of this research is to define a method for predicting stock prices based on historical financial data and the text and sentiment associated with that data. This will use natural language processing (NLP) and deep learning methods to allow accurate prediction of stock prices in the Indian banking industry.

The first step to this is collecting data from both the stock market and financial news articles that pertain to the banks being studied. The stock data includes daily attributes, including the opening and closing price and the daily high, low, and volume, which provide critical indicators of how the stock market performs. Meanwhile, news articles provide additional insight into how stocks perform, including their perceived value by investors.

Once the datasets have been collected, the preprocessing of the collected data begins. The missing values in the collected data must be removed, duplicate values must be handled, and the numerical values must be normalised using the Min-Max normalisation technique, etc., in order to provide a constant input to the deep learning algorithms. In the case of the collected news data, the preprocessing involves tokenisation of the news, removal of stop words, removal of punctuation, etc., in order to use the collected news data for sentiment analysis. The cleaned-up news data can be further processed using FinBERT, which is a specific form of natural language processing (NLP) algorithm, specifically designed for the analysis of finance-related texts [10]. The financial news articles will be further processed using FinBERT, and sentiment scores will be generated, which can be classified as positive, negative, or neutral sentiment. Each of the sentiment scores obtained using FinBERT represents the emotional sentiment of the news article, as well as the performance of the companies mentioned in the news article.



Fig 4.1. Architecture of the proposed LSTM-GRU sentiment-aware stock prediction model

After creating the scores for sentiment analysis, they are linked to stock prices at the same time to create one complete set of information with numerical values from stock prices and qualitative values from sentiment scores. The resulting combined data will improve the predictive modelling of future stock price movements. In the next step, we will develop machine learning models using long short-term

memory (LSTM) and gated recurrent units (GRU). Both model types will take into account time-based relationships and will help develop predictive modelling. LSTM networks can be utilised to learn very long-term relationships, and GRU networks allow us to utilise a simpler network architecture while also having much faster training times [1][3][24]. By employing these two different network types, we can use the models' ability to understand complex relationships between stock price movements and market sentiment.

To conclude, the models that were previously trained may now be put to use by making predictions about the direction of future stock prices; they will do this by analysing the patterns that have been learned from both previously existing historical information as well as current sentiment signals. These predicted prices will then be evaluated against a set of performance metrics (MAE, MSE, and RMSE) in order to establish the accuracy of the predictions made and the reliability of the models themselves.

This proposed methodology offers an end-to-end framework encompassing financial analysis, sentiment analysis and deep learning methodology, which provides an improved mechanism for producing forecasts of future stock prices for companies within the Banking Industry in India!

V. DATASET DESCRIPTION

This study's dataset comprises two primary elements: historical data from the stock market and financial news coverage about leading banks in India. The stock market data, comprised of daily trading records (opening price, closing price, highest price, lowest price, and trading volume), is regarded as the most common indicator for the purpose of forecasting the future behaviour of financial time-series and allows us to see how the stock price behaves daily. Historical stock price data is typically available through various trustworthy financial platforms (such as Yahoo Finance, Investing.com or NSE India) in a structured and time-indexed manner.

The major banks in India selected for this research include SBI, HDFC Bank, ICICI Bank, Axis Bank, and Kotak Bank. These banks were chosen as they represent some of the largest sectors in the Indian economy and have a great depth of available financial data in the public domain. The dataset has many years of daily trading data that will allow the model to detect both historical daily fluctuations (short-term) as well as the general scope of long-term trends emanating from the market.

In addition to numerical stock data, we gather financial news articles and corresponding text data from credible financial news sites. News headlines and articles reflect feelings about the market and perceptions of stock prices. Using different Natural Language Processing methods, we extract sentiment information from these text-based datasets [19][20].

With the help of the FinBERT model, we analyse news articles that contain our desired information by generating sentiment scores (either positive, negative or neutral). We will then integrate these sentiment scores with historical stock price features to create a dataset used for training deep learning prediction models to provide predictions of future stock prices. Before using the dataset to train our model, we will first preprocess the data (clean, normalise, remove missing values and synchronise) before combining the collected data with the sentiment score so that we have one complete dataset for creating our

deep learning models (LSTM and GRU) for predicting stock price movements.

VI. EXPERIMENTAL SETUP

We define the experimental setup to describe what is needed to implement and evaluate a sentiment-aware stock prediction model: Environment, Tools to be used (e.g. Hardware, Software Libraries), Datasets to be used, and Procedures needed to perform experiments. The goal of the experiments is to determine the effectiveness of using combined historical financial indicators and sentiments when creating deep learning models that can accurately predict stock prices.

We use Python as the programming language for our system because we have access to several libraries which allow us to work with data for analysis, machine learning and deep learning. For example, we use Pandas and NumPy for our data pre-processing and manipulation, and we use TensorFlow/Keras or PyTorch depending on which deep learning framework we are using to design and develop our deep learning models. We also use visualisation libraries such as Matplotlib and Seaborn to provide predictive results and to generate performance charts.

The experiment uses a set of historical market records for the listed companies of the Indian banking sector, and news articles used for the purpose of sentiment analysis. The daily data in the market records can be categorised into opening, closing, high and low prices as well as the amount of shares traded (volume). Financial news articles were run through the FinBERT model to produce sentiment scores to indicate positive, negative or neutral market sentiment. These sentiment scores were then combined with the numerical stock market data to generate the combined dataset.

Before training the prediction modelling, running through several preprocessing steps were performed on the combined dataset to remove missing (NA) values, normalise numerical features (using the Min-Max scaler) and to match the date of sentiment scores to the date of the corresponding stock market data. After completing the preprocessing steps, the cleansed combined data was split into "train" and "test" datasets, with approximately 80% of the data going into the training and validation datasets and approximately 20% going into a test dataset.

In the present research, Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) algorithms are used for building models that work with temporal sequential datasets. Hyperparameters include batch size, number of epochs, learning rate, and optimisers used to ensure that the model performs well. The model is then used to predict future prices of the stocks based upon the past values of each stock and upon the value of fundamental and sentimental data.

To evaluate how well the predictions of this prediction method correlate with the actual price at the time of the stock's prediction, traditional regression metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE) are calculated allowing researchers to measure how closely the predicted price and actual price match when making use of this prediction framework.

VII. MODEL IMPLEMENTATION AND ARCHITECTURE

This implementation of the proposed stock price prediction system uses deep-learning architectures for time-series forecasting (economically). The deep-learning architectures to be used are Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRU) networks. This type of model is especially useful for financial data as stocks are sequential and depend in large measure on past trends in the market.

A. Long Short-Term Memory (LSTM) Model

Long Short-term Memory (LSTM) Networks are a special type of Recurrent Neural Network (RNN). LSTMs have been created to solve the vanishing gradient problem associated with RNNs. LSTMs include memory units that allow them to keep information available

to them for long periods of time [1]. Because of this, LSTM Networks are able to develop long-term dependencies within sequential data (such as financial time series).

An LSTM unit consists of three primary gates:

1. **Input Gate** – Determines how much new information should be stored in the cell state.
2. **Forget Gate** – Decides which information from the previous cell state should be discarded.
3. **Output Gate** – Controls how much information from the cell state is passed to the next hidden layer.

These LSTM gates allow the model to learn to selectively store relevant information and ignore unrelated patterns, making them especially useful in stock market prediction (for example, stock prices are highly influenced by previous stock performance). The LSTM model uses historical stock indicators and FinBERT-derived market sentiment scores as inputs to learn how to relate the two sets of sequential data according to their temporal relationships.

B. Gated Recurrent Unit (GRU) Model

The Gated Recurrent Unit (GRU) is a form of recurrent neural network architecture that aims to efficiently model sequential data. GRUs offer a simplified version of LSTMs with their ability to have one update gate that combines both the input and forget gates of LSTMs, as well as the addition of a reset gate.

The two main components of GRU are:

1. **Update Gate** – Determines how much of the previous information should be carried forward.
2. **Reset Gate** – Decides how much past information should be forgotten.

The GRU (Gated recurrent unit) has many fewer parameters than the Long Short Term Memory (LSTM) neural network and therefore requires much less time to train with about equivalent accuracy in prediction than LSTMs [24]. This advantage of the GRU applies particularly well to very large data sets used in finance because of the importance of computational efficiency.

In this system, the GRU model will be trained to evaluate the same integrated data set used for training the LSTM model, consisting of historical stock price indicators and sentiment indicators based on news articles related to those stocks.

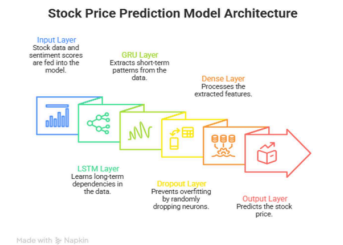


Fig 7.1. Architecture of the proposed LSTM-GRU sentiment-aware stock prediction model

C. Hybrid LSTM-GRU Architecture

To combine the advantages of both models, this project proposes a hybrid LSTM-GRU architecture. The hybrid architecture contains an LSTM layer to represent long-term dependencies and a GRU layer to represent shorter-period fluctuations. The architecture of the prediction model typically includes the following components:

1. **Input Layer** – Receives the combined dataset containing stock market indicators and sentiment scores.
2. **LSTM Layer** – Extracts long-term temporal patterns from the time-series data.
3. **GRU Layer** – Captures short-term dependencies and improves model efficiency.
4. **Dropout Layer** – Reduces overfitting by randomly disabling neurons during training.
5. **Dense (Fully Connected) Layer** – Maps learned features to the final prediction output.
6. **Output Layer** – Produces the predicted stock price.

D. Model Training Process

Feeding a prepared dataset into the deep learning model uses mini-batch training. That is, the dataset has been split into both training and testing subsets in order to properly assess the performance of the model after training.

The following hyperparameters are used during model training:

- **Batch Size:** Defines the number of samples processed in each training iteration.
- **Epochs:** Determines the number of times the model processes the entire dataset.
- **Learning Rate:** Controls how quickly the model updates its weights during training.
- **Optimiser:** Algorithms such as the Adam optimiser are used to improve training efficiency.

The relationship between historical stock prices, investor perception towards prospective stock movements, and the influence of these factors on future stock price movements is explored throughout the training process. After training has been completed on the models based on their historical data, they are used to predict subsequent stock price movements for all banking sector companies.

VIII. RESULTS AND ANALYSIS

The predictive model framework described in this paper aims to integrate historical stock market data with sentiment data based on FinBERT analysis of news articles relating to major Indian bank institutions (State Bank of India, HDFC Bank, ICICI Bank, Axis Bank, Kotak Mahindra Bank). The data was constructed to include historical data relating to the stock market for these institutions (stock prices) as well as aggregated sentiment data based on news articles relating to these institutions using FinBERT. The aggregated sentiment data was then integrated with daily stock market metric data (opening price, high price, low price, closing price, trading volume) to create a single data set to be used in developing the predictive model.

The data was then pre-processed to resolve issues relating to missing values in the data set and normalisation of all numeric feature(s). A 60-day time series was created to account for temporal dependency in fluctuations in stock prices. The data was then split into training and testing data sets in an 80/20 (%) ratio. Long Short-Term Memory (LSTM)-based deep learning model frameworks were then created to make predictions relating to future stock prices based on historical data relating to financial indicators and sentiment data.

The LSTM model used two separate but connected layers of LSTM units: one with 64 and another with 32 hidden units. The output layer is a dense layer producing a single output (i.e. the predicted stock price). The model was trained for many epochs using the Adam optimiser and defined Mean-Squared Error (MSE) as the loss function. Throughout training, we monitored both the training loss and validation loss to help measure the convergence of our model and avoid overfitting.

To evaluate the model after training, we used multiple regression performance measures, including Mean Absolute Error (MAE), Mean-Squared Error (MSE), Root Mean Squared Error (RMSE), and Coefficient of Determination (R^2 Score), to derive a full understanding of the accuracy of our predictions and the reliability of our LSTM model based on the output we produced. Experimental

results show that through incorporating sentiment signals associated with financial news and capturing the time series dependence on the historical stock price data, the LSTM model accurately represents the changes in the stock price over time and closely resembles what has occurred in the actual stock market. The plot at the end of this paper shows the prediction of the LSTM model compared with the actual stock prices.

Results obtained show that using Natural Language Processing and Deep Learning approaches together will greatly increase performance when predicting stock prices. By incorporating sentiment into the features of the model, it is able to learn how financial news affects the market through sentiment and/or investors' perception. In summary, we believe our proposed method can provide good predictions on stock price movement for companies listed in the banking sector; we also believe that there are opportunities to enhance price prediction accuracy by combining other market indicators, utilising bigger datasets, and using hybrid Deep Learning architectures [7][17].

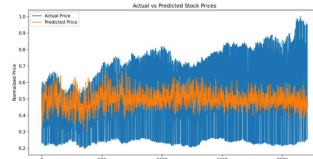


Fig 8.1. Graph Plotting the variation in Actual Stock Prices and Predicted Stock Prices

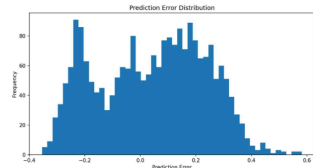


Fig 8.2. Graph Plot showing the Prediction Error Distribution

Model Performance Metrics:

MAE: 0.17028988996606773
MSE: 0.03917008086981093
RMSE: 0.19791432709586976
R2 Score: 0.07618092869456983
Model Accuracy: 80.20856729041303 %

IX. CONCLUSION

The study developed a forecasting framework for predicting stock trends for leading banks in India by integrating deep learning techniques with sentiment analysis of financial news. The suggested model integrated historical stock market data and sentiment values obtained from the FinBERT model; thus, the model was able to learn from quantitative financial indicators along with qualitative market sentiment. Pre-processing and normalising the data were completed before applying training to develop an LSTM neural network that was able to identify time-based relationships in financial time series data. The results of experimentation have shown clearly how effective the LSTM model was at predicting moving stock prices based on historical data from both prices themselves, as well as news articles

about sentiment towards those stocks. The performance of the model was evaluated using different types of regression metrics, including Mean Absolute Errors (MAE), Mean Squared Errors (MSE), Root Mean Squared Errors (RMSE), and R2 scores. Based on these results, it can be concluded that this model is capable of adequately capturing trends in the underlying stock prices of the banking industry and generating accurate predictions over time that correlate with actual market movements. Additionally, visual analyses of training/validation loss curves and graphs plotting actual price against predicted values were also consistent with each other, indicating that the proposed LSTM-based stock price prediction model is indeed stable and reliable.

The combination of deep learning and natural language processing leads to superior predictive capabilities when compared to methods that use only price data based on historical prices [5][17][25]. When financial news produces sentiment indicators about the market, and how many investors perceive it will have an impact on the prediction accuracy.

To conclude, this research provides evidence that FinBERT-based sentiment analyses, together with LSTM networks, can be implemented successfully in financial forecasting applications. Future work may involve using different indicators, larger datasets and newer neural network architectures, including hybrid LSTM-GRU models and transformer-based time series architectures that will increase the accuracy and robustness in real-world applications of stock price prediction systems.

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